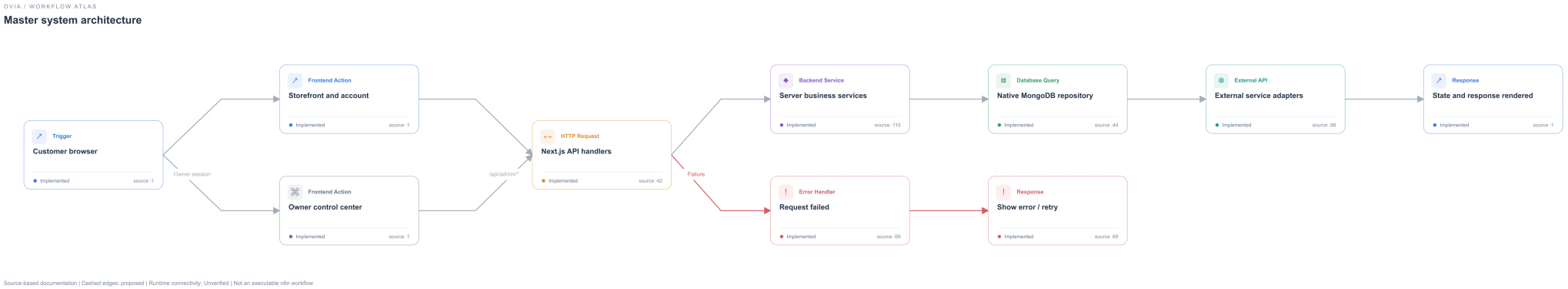
Master system architecture

Architecture overview. Readable node and connection sheets follow.



Master system architecture

Readable detail sheet 1/3. Node numbers and page references preserve every directed connection.

#01 / Trigger / Implemented

Customer browser

Next.js storefront shell with customer-facing pages and client components.

src/app/layout.tsx:1

Next -> #2 (p2); Owner session -> #10 (p4)

#02 / Frontend Action / Implemented

Storefront and account

Catalog, bag, checkout and account state coordinate client requests.

src/components/store-provider.tsx:1

Next -> #3 (p2)

#03 / HTTP Request / Implemented

Next.js API handlers

86 route files, including explicit dispatch branches and provider webhooks.

src/app/api/v1/[[...path]]/route.ts:42

Next -> #4 (p2); Failure -> #8 (p3)

#04 / Backend Service / Implemented

Server business services

Commerce, accounts, logistics, mail, support, marketing and reports.

src/server/commerce.ts:115

Next -> #5 (p3)

Master system architecture

Readable detail sheet 2/3. Node numbers and page references preserve every directed connection.

#05 / Database Query / Implemented

Native MongoDB repository

MongoClient pool; repository access and withTransaction for coordinated writes.

src/server/db.ts:44

Next -> #6 (p3)

#06 / External API / Implemented

External service adapters

Razorpay, PayPal sandbox, Delhivery, Resend, Algolia, OAuth, Gmail, translation and DeepSeek. Runtime configuration and connectivity are Unverified.

src/server/integrations-service.ts:88

Next -> #7 (p3)

#07 / Response / Implemented

State and response rendered

Safe route responses update customer and owner views.

src/components/store-provider.tsx:1

END: no outgoing connection

#08 / Error Handler / Implemented

Request failed

Validation, access, conflict or service errors are returned to the caller. Status and error shape depend on the specific route.

src/server/http.ts:69

Next -> #9 (p4)

Master system architecture

Readable detail sheet 3/3. Node numbers and page references preserve every directed connection.

#09 / Response / Implemented

Show error / retry

Failure end state. Caller receives the actual error; a retry may be offered by the UI.

src/server/http.ts:69

END: no outgoing connection

#10 / Frontend Action / Implemented

Owner control center

Server-protected owner operations and API-specific authorization.

src/components/admin/admin-shell.tsx:1

/api/admin/* -> #3 (p2)

Master system architecture

Complete directed connection register / no omitted success, failure or proposed edges

#01 Customer browser -> #02 Storefront and account | Next

#02 Storefront and account -> #03 Next.js API handlers | Next

#03 Next.js API handlers -> #04 Server business services | Next

#04 Server business services -> #05 Native MongoDB repository | Next

#05 Native MongoDB repository -> #06 External service adapters | Next

#06 External service adapters -> #07 State and response rendered | Next

#03 Next.js API handlers -> #08 Request failed | Failure | FAILURE

#08 Request failed -> #09 Show error / retry | Next | FAILURE

#01 Customer browser -> #10 Owner control center | Owner session

#10 Owner control center -> #03 Next.js API handlers | /api/admin/*

Note: Source implementation is documented; deployed configuration and provider connectivity are not verified.

Note: MongoDB Atlas deployment topology is Unverified. The code proves a MongoDB connection and database selection, not specific cluster tiers or regions.